Input
unit_2d_down(c_in)
unit_2d_keep(c_out)
unit_3d(c_out)
Output
$H_{in} \times W_{in} \times C_{in}$
$H_{out} \times W_{out} \times C_{out}$
$H_{out} \times W_{out} \times C_{out}$
$H_{out} \times W_{out} \times C_{out}$